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Baum

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(54) **SURFACE WATER BIOSOURCE
AUGMENTATION PRODUCTION AND
DISTRIBUTION SYSTEM**

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2019, now Pat. No. 11,840,467.

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C02F 103/00 (2006.01)

(52) **U.S. Cl.**

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(2013.01); **C02F 2103/007** (2013.01); **C02F**
2203/006 (2013.01)

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None

See application file for complete search history.

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Primary Examiner — Chester T Barry

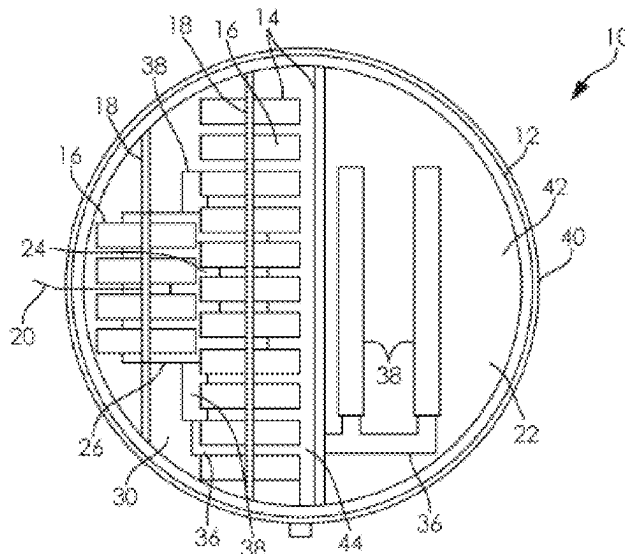
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(57)

ABSTRACT

A modular system for aeration and biosource augmentation
of a body of water and a method of using the system to
prevent or eliminate an algal bloom in a water body include
one or more mechanical basins located outside the body of
water and at least two of: one or more residence pods
submerged in the body of water that aerate the body of water,
dose beneficial biosource, host biosource, and provide bio-
source to the body of water; one or more residence air pods
submerged in the body of water that aerate the body of water,
host beneficial bacteria, and provide beneficial bacteria to
the body of water; and one or more residence nano pods
submerged in the body of water that host, the one or more
residence pods or the one or more residence air pods in fluid
communication with the one or more mechanical basins.

18 Claims, 11 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/630,119, filed on Feb. 13, 2018, provisional application No. 62/775,365, filed on Dec. 4, 2018.

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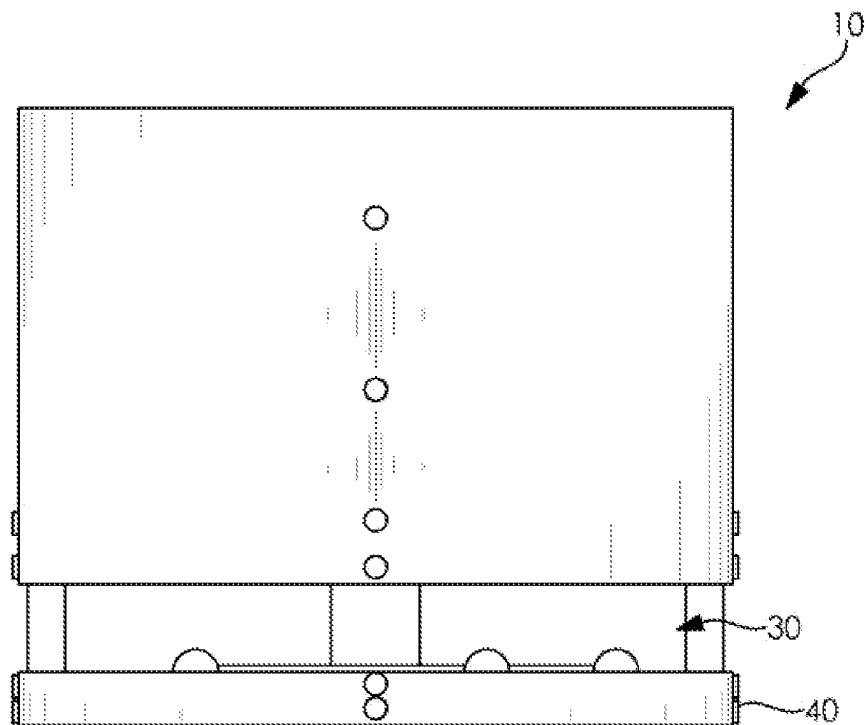


FIG. 1A

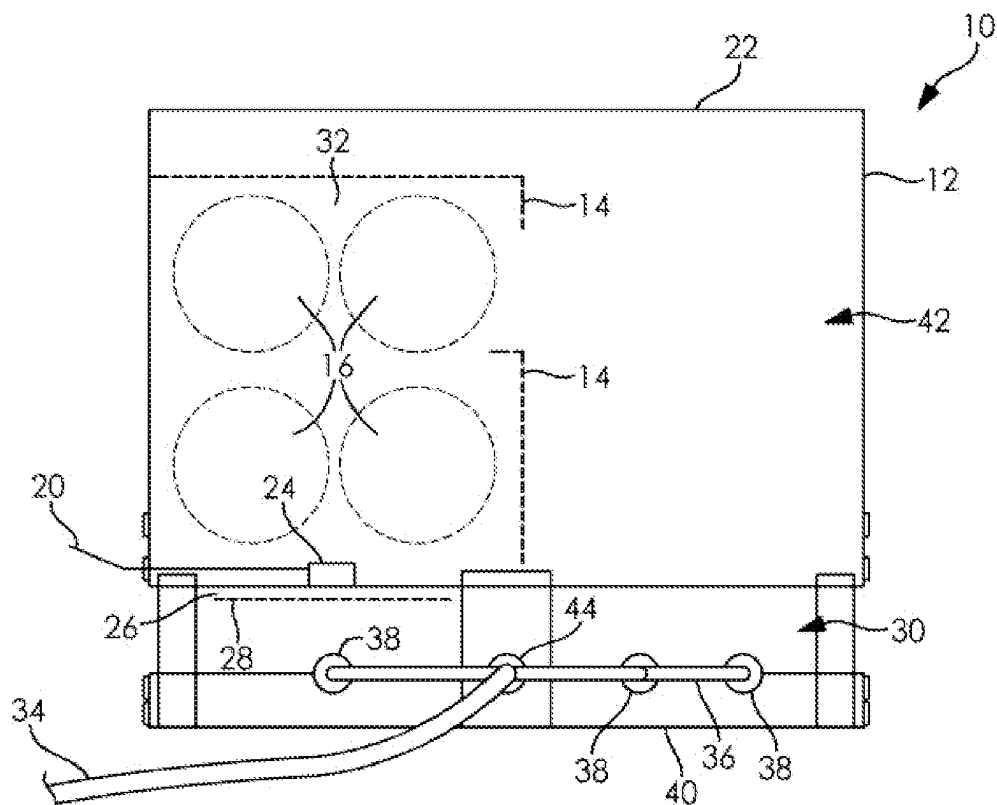
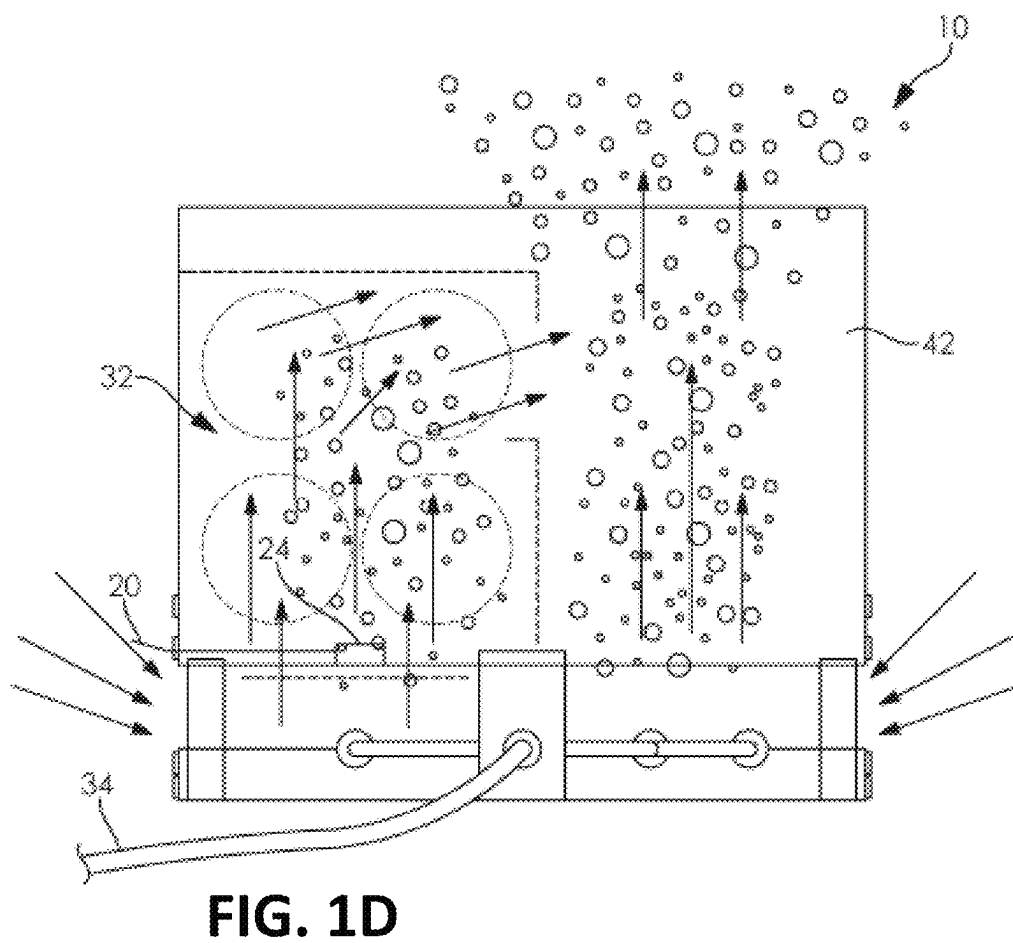
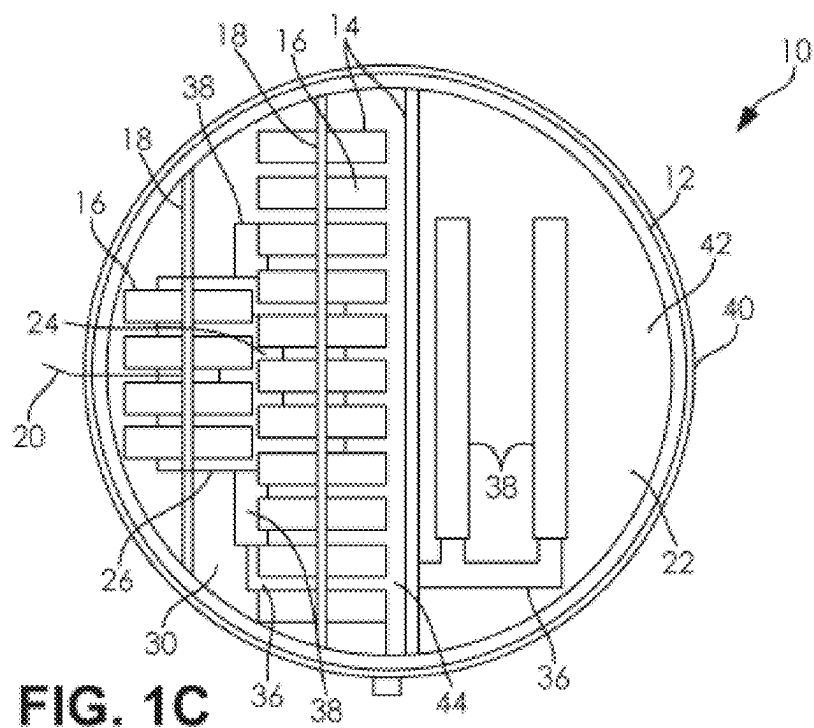


FIG. 1B



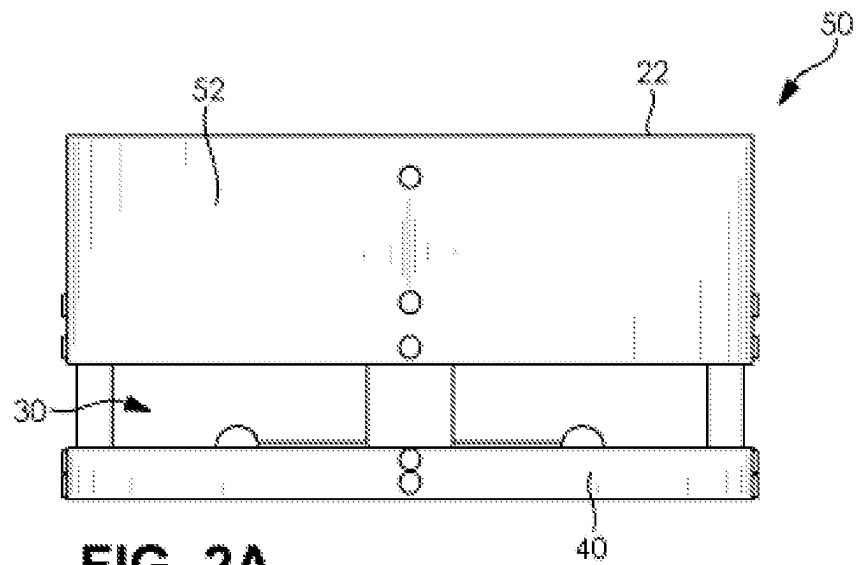


FIG. 2A

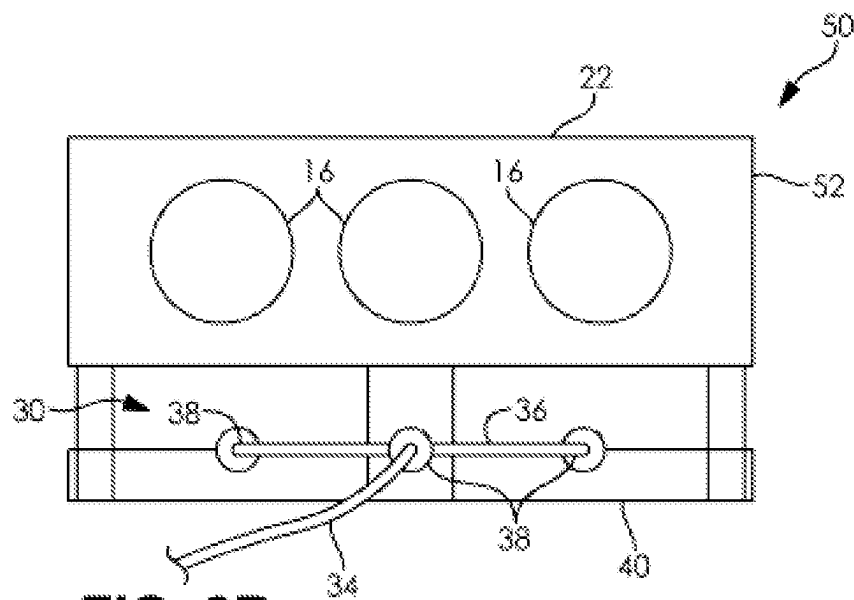
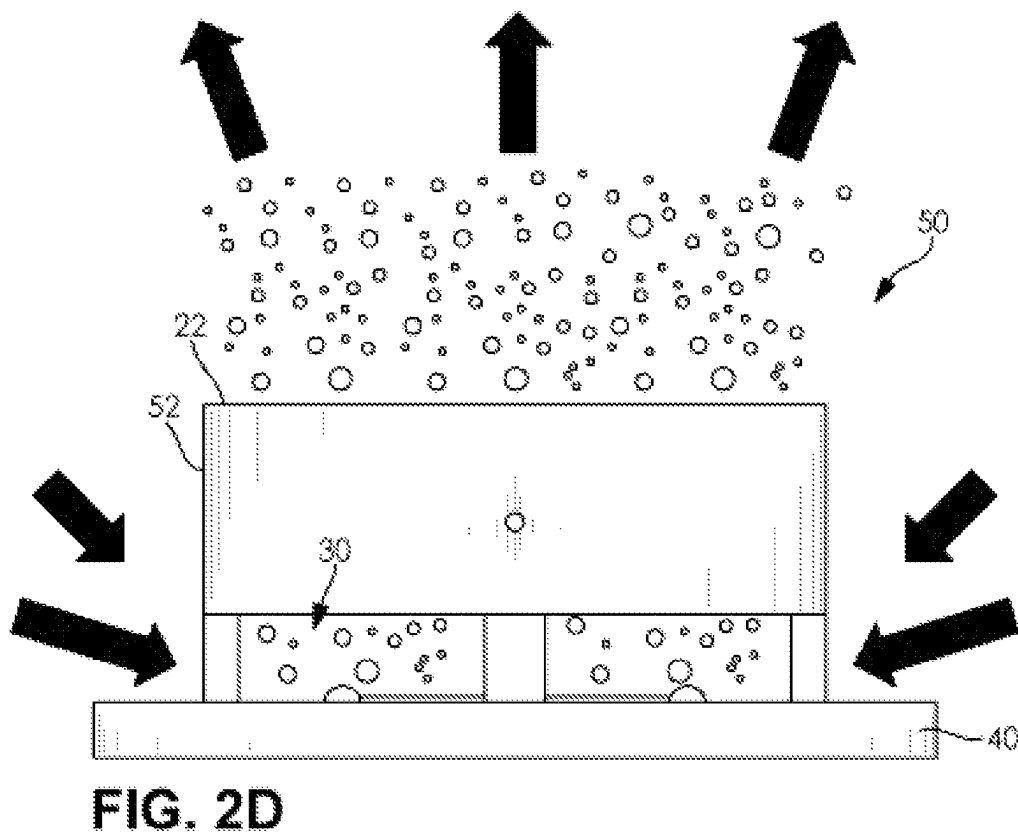
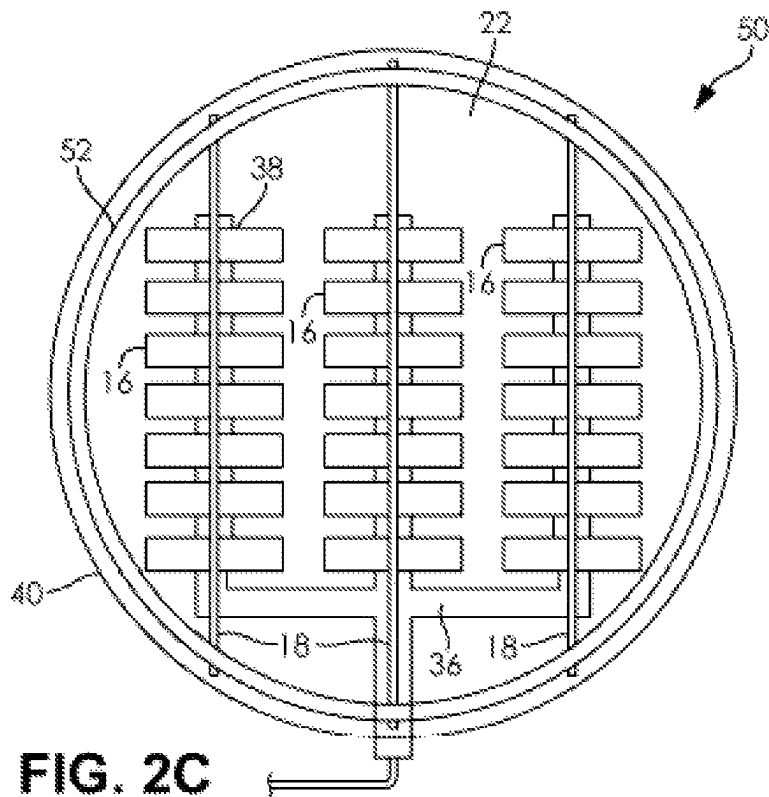
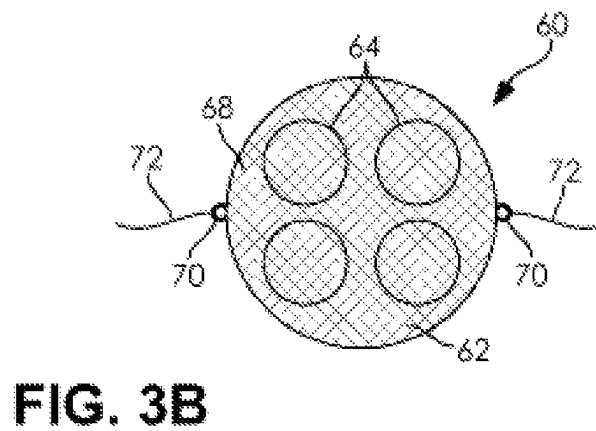
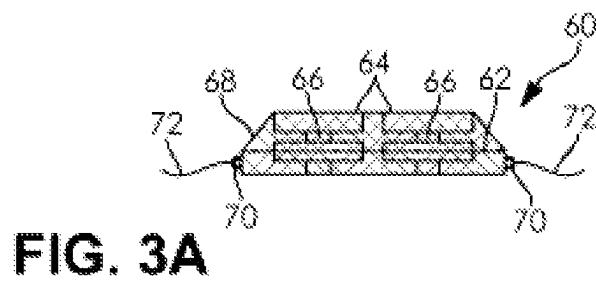
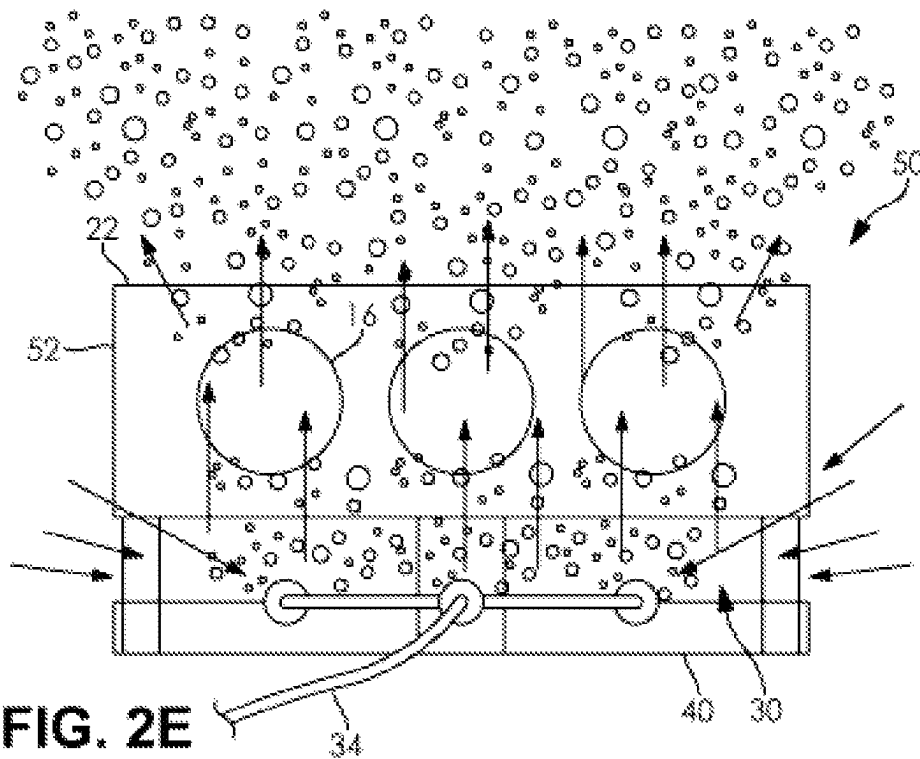


FIG. 2B





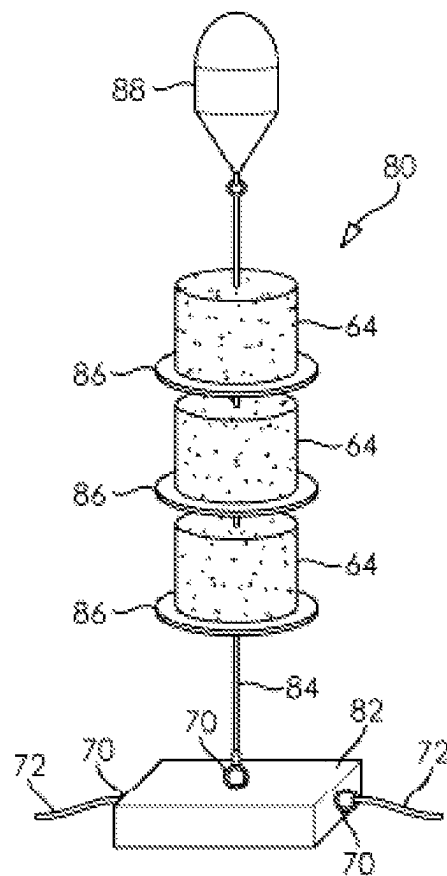


FIG. 4

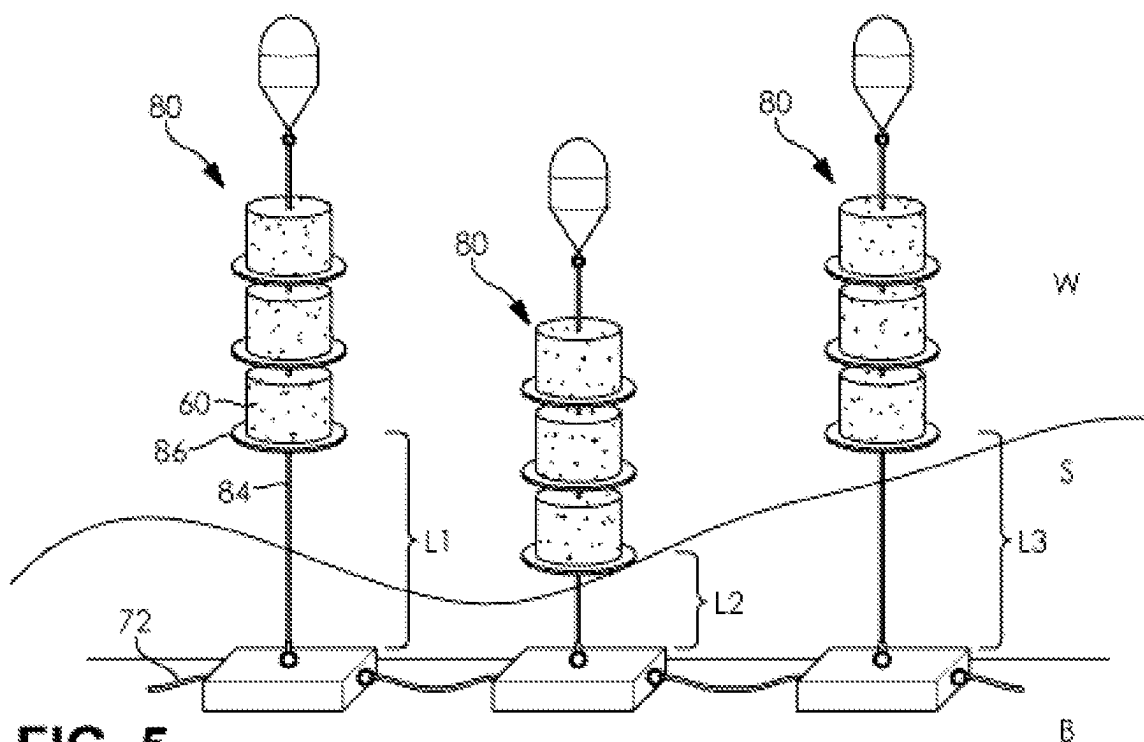


FIG. 5

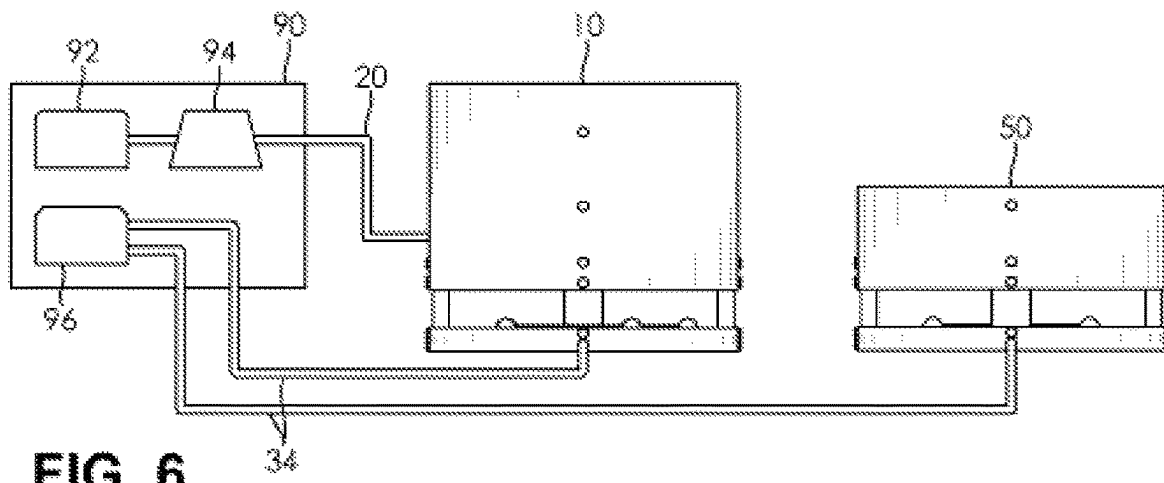


FIG. 6

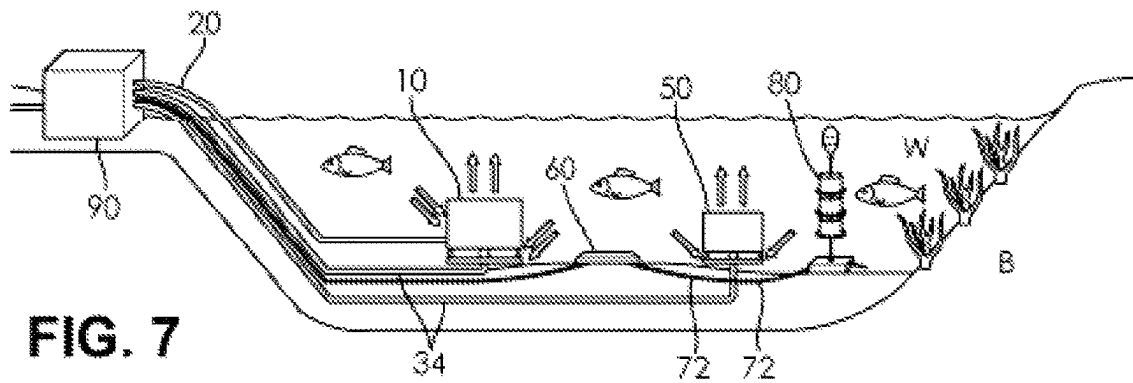


FIG. 7

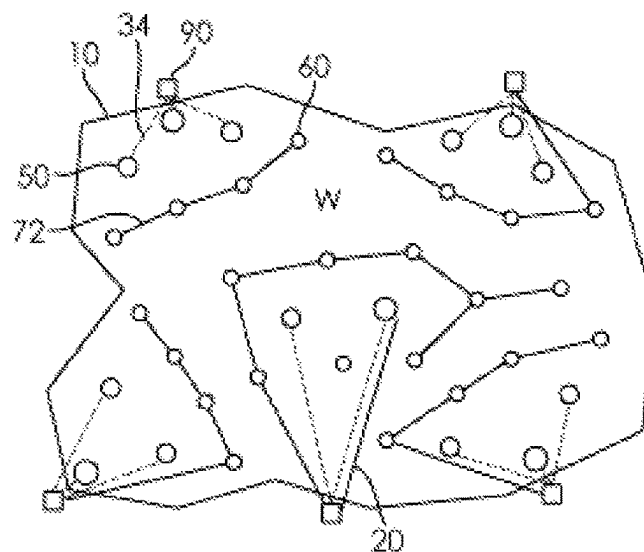
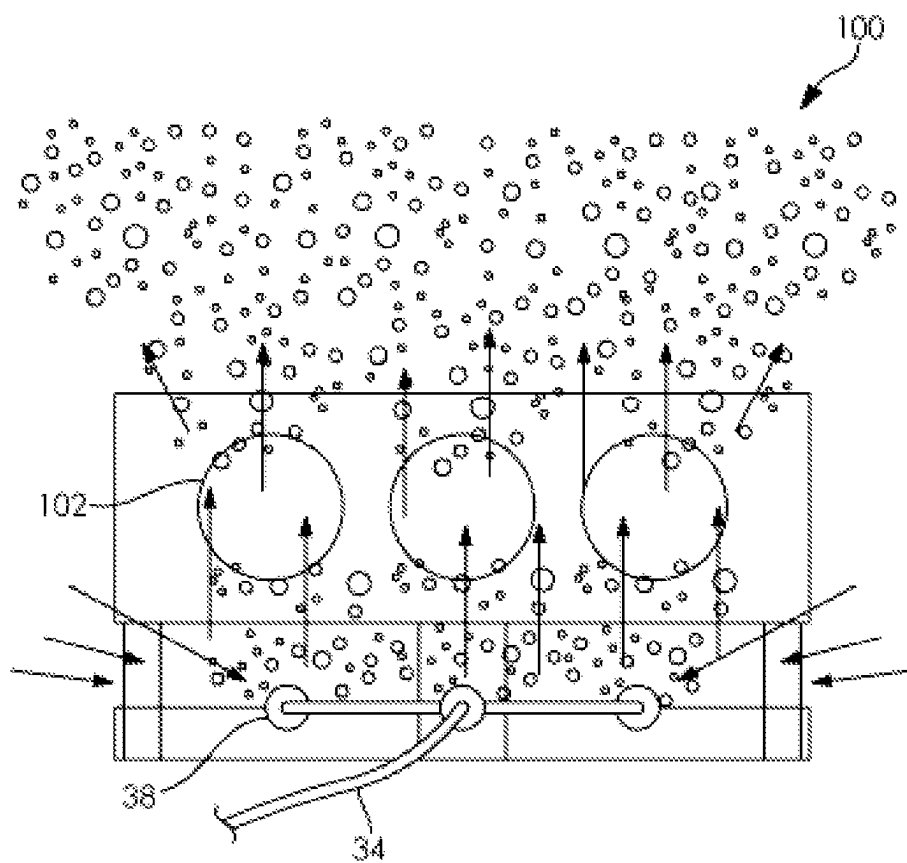


FIG. 8

**FIG. 9**

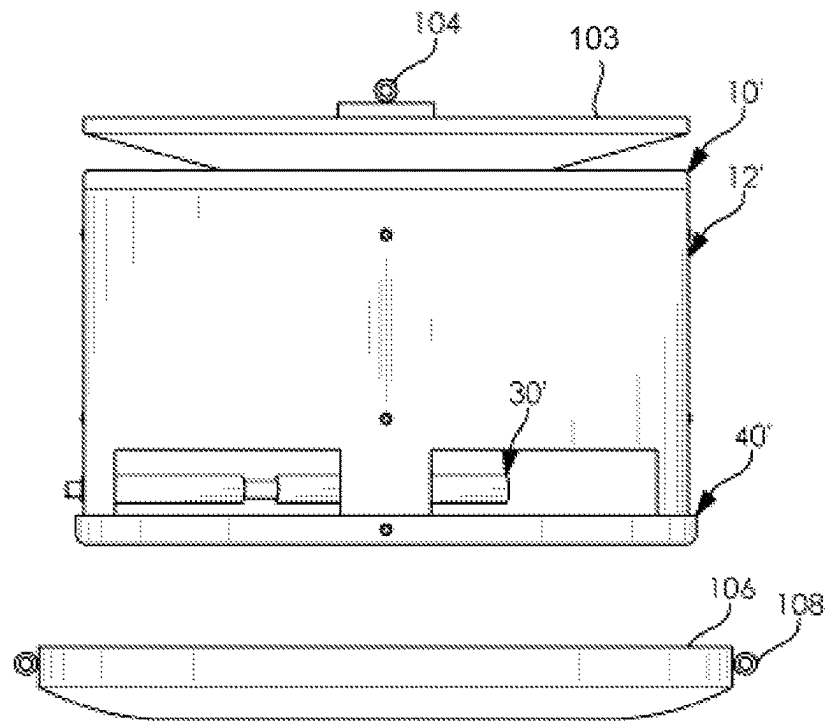


FIG. 10

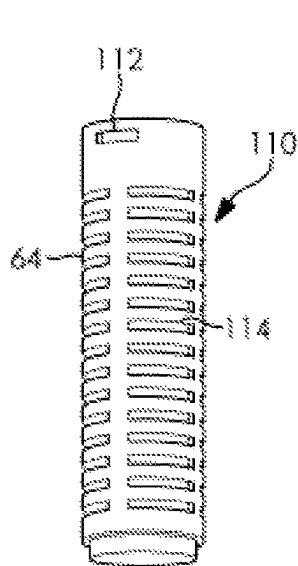


FIG. 11

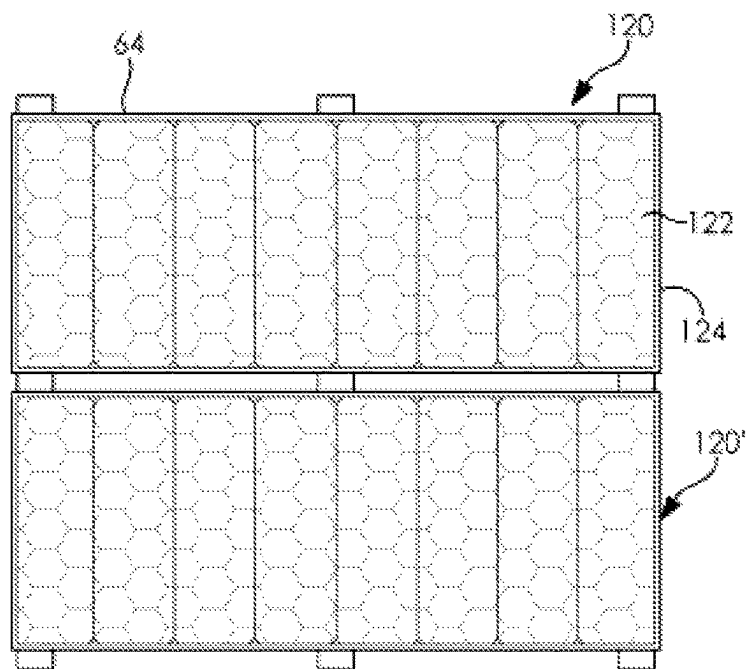


FIG. 12A

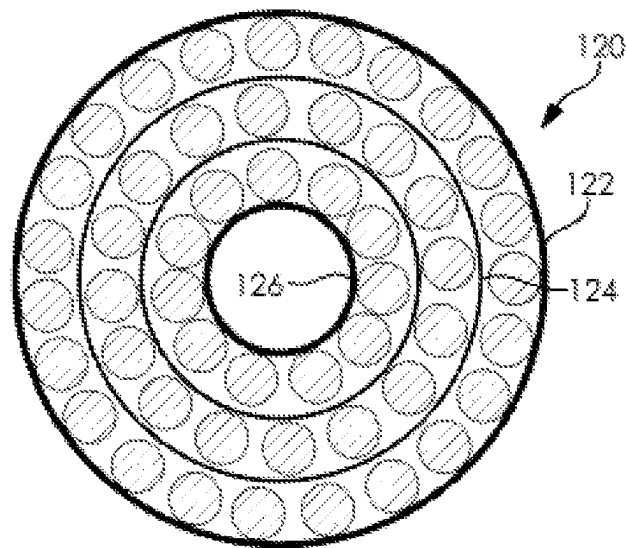


FIG. 12B

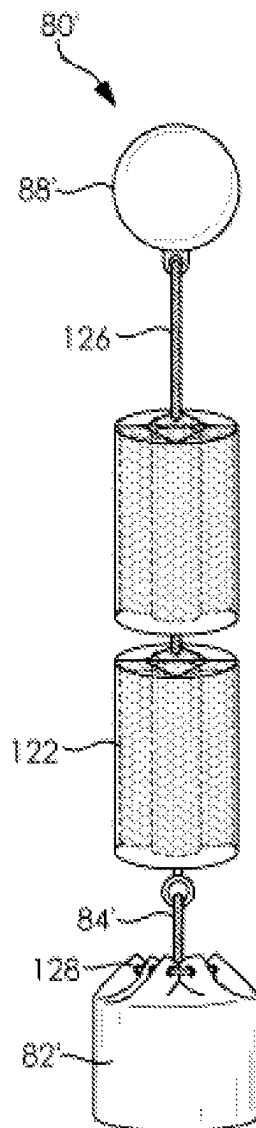


FIG. 13A

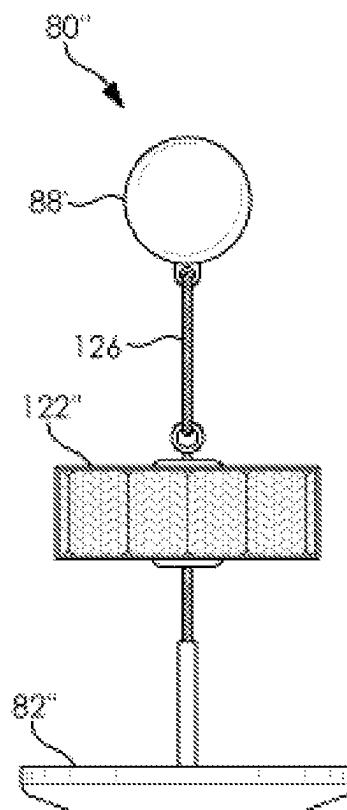


FIG. 13B

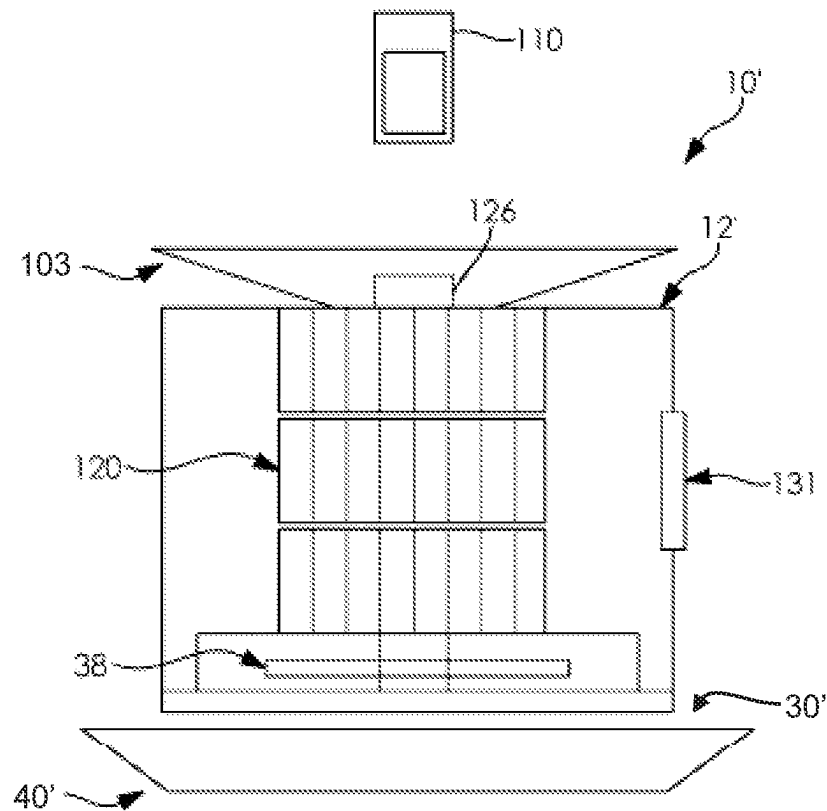


FIG. 14A

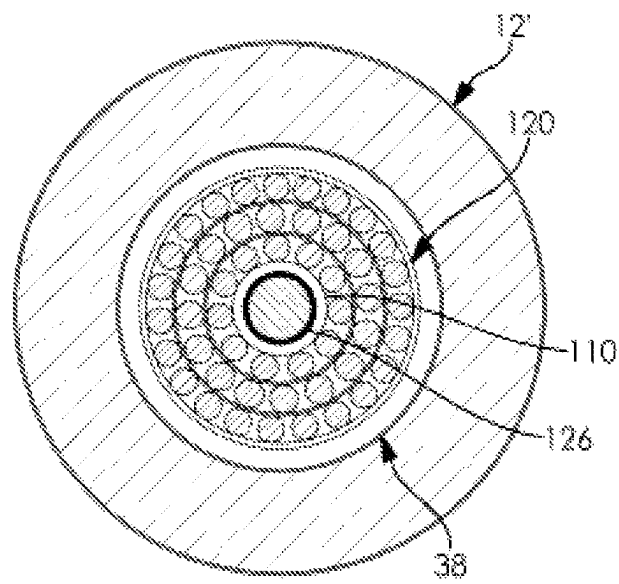


FIG. 14B

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SURFACE WATER BIOSOURCE AUGMENTATION PRODUCTION AND DISTRIBUTION SYSTEM

RELATED APPLICATIONS

This application is a continuation U.S. patent application Ser. No. 16/969,295 filed Aug. 12, 2020; now U.S. Pat. No. 11,840,467 B2; which in turn is a US National Phase filing of PCT/US2019/017590 filed Feb. 12, 2019; which in turn claims priority benefit of U.S. Provisional Application Ser. No. 62/630,119 filed Feb. 13, 2018 and U.S. Provisional Application Ser. No. 62/775,365 filed Dec. 4, 2018; the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

The present invention generally relates to the field of water treatment, and more specifically to a system and method for biosource augmentation and aeration of standing bodies of water such as ponds, lakes, or waste water reservoirs.

BACKGROUND

Algal bloom is a rapid increase or accumulation in the population of algae in freshwater or marine water systems. Algal blooms are characterized by the discoloration in the water sources resulting from pigments released by the algae. Freshwater algal blooms typically result from an excess of nutrients such as phosphates that originate from fertilizers that are applied to land for agricultural or recreational purposes, and may also originate from household cleaning products containing phosphorus. When phosphates are introduced into water systems, the higher nutrient concentrations cause increased growth of algae and plants. In addition excess carbon and nitrogen are also suspected as causes of algal blooms. The presence of residual sodium carbonate also acts as catalyst for the algae to bloom by providing dissolved carbon dioxide for enhanced photosynthesis in the presence of nutrients. The excess nutrients that support algal blooms enter watersheds through water runoff.

As algal blooms grow, the algal blooms deplete the oxygen in the water and block sunlight from reaching fish and plants. Algal blooms can last from a few days to many months. With less light, plants beneath the bloom can die and fish can starve. While algae tend to grow very quickly under high nutrient availability, each alga is short-lived, and the result is a high concentration of dead organic matter which starts to decay and form a muck. The microbes which decompose the dead algae use up even more dissolved oxygen in the water, resulting in hypoxic conditions, which in turn causes more fish to die or leave the area. Without sufficient dissolved oxygen in the water, animals and plants continue to die off in large numbers.

Man-made efforts to control an outbreak of algal bloom have included the use of algaecides for killing algae, and have been used mostly in small bodies of water. For large algal blooms, however, adding algaecides such as silver nitrate or copper sulfate can have worse effects, such as killing fish outright and harming other wildlife. The negative effects can therefore be worse than letting the algae die off naturally.

In addition, biosource augmentation, bacteria grow/filtration media, and aeration have been used in lakes, ponds, and wastewater treatment to combat algal blooms. Bacterial augmentation typically involves the application of a blend of

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beneficial bacteria to the water. There are numerous blends of various strains of bacteria available in a variety of forms including, liquid, powder, pellets, disks, and others. There are also systems that promote growth of native bacteria already present in the water, as well as the commercially produced blends of beneficial bacteria. Bacterial grow/filtration media is most commonly found in fish ponds and waste water treatment applications. Growth media provides a place for beneficial bacteria cultures to live and produce. Aeration is another key component in pond and lake management for prevention of algae build up. Aeration systems increase dissolved oxygen levels in treated water and in so doing promote healthy aquatic life and improved water quality, as well as promoting the growth of beneficial bacteria.

While there are available methods for management of bacterial growth and prevention of algal blooms in ponds and lakes, there continues to be a need for methods and systems that improve biosource augmentation treatments in lakes and ponds. There also exists a need for a system to reduce decomposing organic material muck in surface water bodies such as lakes and ponds.

SUMMARY

A modular system for aeration and biosource augmentation of a body of water is provided. According to embodiments the modular system includes one or more mechanical basins located outside the body of water and at least two of: one or more residence pods submerged in the body of water that aerate the body of water, dose beneficial biosource, host biosource, and provide biosource to the body of water; one or more residence air pods submerged in the body of water that aerate the body of water, host beneficial bacteria, and provide beneficial bacteria to the body of water; and one or more residence nano pods submerged in the body of water that host, the one or more residence pods or the one or more residence air pods in fluid communication with the one or more mechanical basins.

A method of using a modular system for aeration and biosource augmentation of a body of water to prevent or eliminate an algal bloom in a water body is also provided. According to embodiments, the method includes positioning one or more mechanical basins along the outside of the water body, connecting one or more residence pods to the one or more mechanical basins with a first set of one or more tubes for supplying air and a second set of one or more tubes for supplying biosource to the residence pods, connecting the one or more residence air pods to the one or more mechanical basins with the first set of one or more tubes for supplying air to the residence air pods, and submerging the one or more residence pods and the one or more residence air pods in the body of water.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is further detailed with respect to the following drawings that are intended to show certain aspects of the present invention, but should not be construed as a limit on the practice of the invention, wherein:

FIG. 1A is a side view of a residence pod in accordance with embodiments of the invention;

FIG. 1B is a side cutaway view of the residence pod shown in FIG. 1A in accordance with embodiments of the invention;

FIG. 1C is a top view of the residence pod shown in FIG. 1A in accordance with embodiments of the invention;

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FIG. 1D is a side cutaway view of the residence pod shown in FIG. 1A showing water flow and air flow through the residence pod in accordance with embodiments of the invention;

FIG. 2A is a side view of a residence air pod in accordance with embodiments of the invention;

FIG. 2B is a side cutaway view of the residence air pod shown in FIG. 2A in accordance with embodiments of the invention;

FIG. 2C is a top view of the residence air pod shown in FIG. 2A in accordance with embodiments of the invention;

FIG. 2D is a side view of the residence air pod shown in FIG. 2A showing water flow and air flow in the form of bubbles through the residence air pod in accordance with embodiments of the invention;

FIG. 2E is a side cut away view of the residence pod shown in FIG. 2A showing water flow and air flow in the form of bubbles through the residence air pod in accordance with embodiments of the invention;

FIG. 3A is a side sectioned view of a residence nano pod in accordance with embodiments of the invention;

FIG. 3B is a top view of a residence nano pod in accordance with embodiments of the invention;

FIG. 4 is a side perspective view of a residence stringer pod in accordance with embodiments of the invention;

FIG. 5 illustrates the deployment of a series of residence stringer pods of FIG. 4 in a body of water in accordance with embodiments of the invention;

FIG. 6 is a functional block diagram of a residence pod system that illustrates the connections between pumps for air and beneficial biosources and a residence pod and a residence air pod in accordance with embodiments of the invention;

FIG. 7 is a cross sectional view of a residence pod system deployed in a pond or lake in accordance with embodiments of the invention;

FIG. 8 is a top view of a deployment layout of a residence pod system in a pond or lake in accordance with embodiments of the invention;

FIG. 9 is a side sectioned view of a residence pod with time release containers of biosource in accordance with embodiments of the invention;

FIG. 10 is a side view of a residence pod in accordance with another embodiment of the invention;

FIG. 11 is a perspective view a biosource cartridge;

FIGS. 12A and 12B are schematics of a circular array of pods for containing bacteria growth media, in side view stacked in modular form with a duplicate array (FIG. 12A), and a top view of the array (FIG. 12B);

FIG. 13A is a side perspective view of a residence stringer pod in accordance with another embodiment of the invention;

FIG. 13B is a side perspective view of a residence stringer pod in accordance with still another embodiment of the invention;

FIG. 14A is a partial cutaway side view of the residence pod of FIG. 10; and

FIG. 14B is a top view of the residence pod of FIG. 14A with the deflector lid removed for visual clarity.

DETAILED DESCRIPTION OF THE INVENTION

The present invention has utility as a modular system and method that improves aeration and biosource augmentation treatments in lakes and ponds. In a specific inventive embodiment the system includes three units: a biosource

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dosing, hosting, aeration, and distribution unit herein referred to as a “residence pod;” a biosource hosting, aeration, and distribution unit referred to herein as a “residence air pod;” and a biosource hosting and distribution unit referred to as a “residence nano pod.” Inventive embodiments of the modular treatment systems may be installed in lakes and ponds to provide an ongoing supply of commercially available blends of biosource designed to effectively consume and digest organic material “muck” that builds up on the bottom of these bodies of water. Additionally, embodiments of the inventive modular treatment system insure a constant supply of beneficial bacterial that consume and digest available nutrients in the body of water.

As used herein, a “biosource” is defined as microorganisms, insect larvae, enzymes, nutrients to promote bacterial growth, or a combination thereof capable of digesting organic muck sediment. In some embodiments, a microorganism is aerobic bacteria that thrive in the aqueous environment. It is appreciated that the surface water defining the aqueous environment is fresh water, brackish, or salt water.

As used herein, the terms “pod” and “pod unit” are used synonymously.

Embodiments of the inventive residence pod unit incorporate aeration, biosource augmentation, and grow/filtration media. A residence pod unit is connected to a mechanical basin containing at least an air compressor to provide aeration fluid communication to other system components. In still other inventive embodiments, the mechanical basin has a liquid biosource reservoir, nutrient reservoir, or combination thereof in fluid communication with a doser pump that is located out of the body of water being treated to meter surface water active organisms to the other system components. The air compressor delivers a continuous flow of air through tubing to one or more diffusers. The doser pump delivers metered quantities of the biosource through tubing to a dose receptacle. The organism metering being automatic and continuous, on set intervals, or based on a preselected program as to timing and amounts delivered. A set of air diffusers that are located near the bottom of the residence pod unit generate a continuous supply of small bubbles that move upward through a flow chamber drawing water through a bottom water intake and out through the top. Another diffuser located under the grow chamber generates an ongoing supply of small bubbles traveling upward, providing oxygen for growth of bacteria and also drawing nutrient rich water that provides food for bacteria. This nutrient rich water full of bacteria flows from the grow chamber into the flow chamber where it joins the flow upward, out of the unit, and into the open water. The residence pod unit continuously produces and distributes the bacteria, some of which may consume nutrients in a water column of the body of water while other bacteria may consume muck build up on the bottom of the body of water.

Embodiments of the inventive residence air pod unit incorporate aeration, and grow/filtration media. In embodiments of the residence air pod unit, the grow media provides an ideal place for both native beneficial bacteria as well as the biosource, such as bacteria traveling from other residence pods that are in the vicinity to grow. A mechanical basin containing an air compressor delivers continuous flow via tubing to one or more diffusers. The diffuser(s) are located near the bottom of the residence air pod unit and produce small bubbles that move upward and discharge from the residence air pod unit at the top. These bubbles deliver oxygen that promote bacterial growth in the grow media while creating a flow through of water. As this water flows through the residence air pod unit biosource treats the

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water. This treated water, full of biosource, flows into the open water. The residence air pod unit continuously produces and distributes the biosource, some of which consumes nutrients in the water column of the body of water, while other beneficial bacteria consume muck build up on the bottom of the body of water. The combining of aeration and grow media into a single unit is a significant innovation not found in prior algal treatment solutions.

Embodiments of the inventive residence nano pod unit have a base with grow/filtration media. The inventive residence nano pod units reside underwater and provide a place for native beneficial bacteria or beneficial bacteria, or other forms of biosource, if present, from other residence pods, residence nano pod units, or a combination thereof to grow. Natural water movement provides some oxygen and flow through the residence nano pod units that result in distribution of the beneficial bacteria into the surrounding area. Each unit has a mechanical fastener, such as an eye bolt for attaching cables and connecting strings of units together. Such fasteners are readily supplied on two sides of a given pod unit. One cable extends from the string to the shore and out of the lake. This cable system provides a way to locate and service residence nano pods by simply following the cables.

Each of the aforementioned types of pods, each individual pod independently is weighted to reside on the bottom of the body of water, tethered to the bottom and has buoyancy, is mounted to a pole or other substrate in a submerged position, or suspended in a submerged position from a buoy. It is appreciated that an inventive system include different types of suspension or bottom positions pods functioning together in a single inventive system.

In inventive embodiments of the residence pod system, each of the different units (residence pod, residence air pod, residence nano pod) may work independently and provide treatment benefits to a body of water illustratively including a lake or pond. However, it is the combined use of all the units together that result in a broadest and most effective and powerful impact on reducing muck and nutrient levels in a body of water. Embodiments of the inventive residence pods produce and distribute the greatest amount of biosource that treats the water and find their way to other residence units. Embodiments of the inventive residence air pods also host, grow, and distribute, beneficial bacteria. Embodiments of the residence nano pods are the least powerful of the units, however the residence nano pods are less expensive to produce and may be deployed in greater numbers. Therefore numerous residence nano pods units may be placed in a treatment area cost effectively to increase overall coverage area of the system. Embodiments of the inventive residence pod systems are scalable and custom designed to each application, and may be used in small backyard ponds or lakes, regardless of the degree of salinity. All the residence units have an area of influence and by spacing these units within close enough proximity to other units a very potent and effective biosource augmentation treatment program is achieved. Embodiments of this combined system are a new and innovative approach to biosource augmentation in bodies of water.

Referring now to the figures, FIGS. 1A-1D illustrate an inventive embodiment of a residence pod 10 with a main housing 12, a base 40, and an open water intake 30. As best shown in FIGS. 1B and 1C, the main housing 12 of the residence pod 10 has a partition 14 that separates a grow chamber 32 from a flow chamber 42. The grow chamber 32 has a series of bacterial growth media 16 in the form of disks held by hanger rods 18. The bacterial growth media 16

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provides an ideal place for both native beneficial bacteria as well as the bacteria traveling from other residence pods that are in the vicinity to grow. The disks of bacterial growth media 16 are spaced along the axis of the hanger rods 18 to allow water to flow between the disks 16 as shown by the arrows representing water flow in FIG. 1D. A tube 20 leads into the grow chamber 32 that delivers a liquid biosource blend from a peristaltic metering pump to a bacteria dose receptacle 24 that distributes the delivered biosource, such as bacteria to a bacteria cloth grow media 26 supported by a screen 28 positioned at the water intake 30 at the bottom of the residence pod 10. A flexible air tube 34 connected to an air compressor, located outside of the body of water where the pod 30 is being placed, feeds air into the residence pod 10 via an air control valve 44 and an air distribution tube 36, which may be formed of polyvinyl chloride (PVC), with outlets that supply air to a set of air diffusers 38. In some inventive embodiments, the tube is weighted to assure sitting on bottom. As best shown in FIG. 1D bubbles generated by the air outputted from the set of air diffusers 38 exit the open top outlet 22 of the residence pod 10. The bubbles from the air diffusers 38 move upward through the flow and growth chambers and draw in water at the water intake 30 at the bottom of the residence pod 10. These bubbles deliver oxygen that promote bacteria growth in the grow media while creating a flow through of water. As this water flows through the residence pod unit 10, the water is treated by biosource in the grow chamber 32. This treated water, full of beneficial bacteria and oxygen, flows into the body of water.

FIGS. 2A-2E illustrate an inventive embodiment of a residence air pod 50 with a main housing 52, a base 40, an open water intake 30, and an open water outlet 22. As best shown in FIG. 1B the main housing 52 of the residence air pod 50 encloses a series of bacterial growth media 16 in the form of disks held by hanger rods or brackets 18. The bacterial growth media 16 provides an ideal place for both native beneficial bacteria as well as the bacteria traveling from other residence pods that are in the vicinity to grow. The disks of bacterial growth media 16 are spaced along the axis of the hanger rods or brackets 18 to allow water to flow between the disks 16 as shown by a set of arrows representing water flow in FIG. 2E. A flexible air tube 34 connected to an air compressor, located outside of the body of water in which the residence air pod 50 is placed, feeds air into the residence air pod 50 via an air distribution tube 36, which may be formed of polyvinyl chloride (PVC), with outlets that supply air to a set of air diffusers 38. As best shown in FIGS. 2D and 2E bubbles generated by the air outputted from the set of air diffusers 38 exit the open top outlet 22 of the residence air pod 50. These bubbles deliver oxygen that promote bacteria growth in the grow media while creating a flow through of water. As this water flows through the residence air pod unit 50, the water is treated by beneficial bacteria in the grow chamber 32. This treated water, full of beneficial bacteria and other biosource, if present, and oxygen, flows into the open water.

FIGS. 3A and 3B illustrate an inventive embodiment of a residence nano pod 60. A base tray 62 supports an array of bacterial growth media 64. In a specific inventive embodiment, the bacterial growth media 64 in the array are stacked in columns at least two high and are separated by spacers 66. A netting 68 attached to the base tray 62 encapsulates the bacterial growth media 64. A set of eye bolts 70 connected to opposing edges of the base tray 62 provide an attachment or connection point for a connector cable 72. In a specific

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inventive embodiment a plurality of residence nano pods **60** may be joined as a string of pods that may be laid along the bottom of a lake or pond.

FIG. 4 illustrates an inventive embodiment of a residence stringer pod **80**. The residence stringer pod **80** has a weighted base **82** with an eye bolt **70** extending upward from the center of the weighted base **82**. A variable length extension line **84** has a proximal end secured to the weighted base and a distal end connected to a float **88**. The float **88** pulls up on the extension line **84** toward a surface waterline when the float **88** is submerged under the waterline of a body of water in which the residence stringer pod **80** is placed. One or more media support plates **86** each holding bacterial growth media **64** are concentrically attached along the length of the extension line **84**. The indigenous bacteria and the bacteria from other residence pods find, live, and grow in the growth media. Natural water flow provides oxygen, nutrients, and distributes the biosource. A mechanical fastener such as eye bolts **70** may be connected to opposing edges of the base tray **82** provide an attachment or connection point for a connector cable **72**. In a specific inventive embodiment, a plurality of residence stringer pods **80** may be joined as a string of pods that may be laid along the bottom of a lake or pond as shown in FIG. 5, where three separate residence stringer pods **80** are resting on the hard bottom B of a pond or lake. The variable length (L1, L2, L3) of the extension lines **84** are adjusted so as to position the media support plates **86** that hold the bacterial growth media **64** above the muck layer of sediment S and decayed dead algae and position the bacterial growth media in the water W of the pond or lake, so that the bacteria can grow and distribute in the body of water.

FIG. 6 is a functional block diagram of a residence pod system that illustrates the connections between a mechanical basin **90** and a residence pod **10** and a residence air pod **50**. The mechanical basin **90** houses pumps for air **96** and beneficial biosource in a reservoir **92** distributed with a biosource pump **94** illustratively including a peristaltic metering or doser pump. The biosource pump **94** is connected to the residence pod **10** via tubing **20**, while the air pump supplies air to the air diffusers **38** in the residence pod **10** and residence air pod **50** via flexible air tubes **34**.

FIG. 7 is a cross sectional view of a residence pod system deployed in a pond or lake. The residence pod system shown deployed includes a residence pod **10**, a residence nano pod **60**, a residence air pod **50**, and a residence stringer pod **80**. The residence nano pod **60** and the residence stringer pod **80** are both tethered together and are also tethered to the mechanical basin **90** with connector cable **72**. The mechanical basin supplies air to the residence pod **10** and to the residence air pod **50** via flexible air tubes **34**, and also supplies biosource to the residence pod **10** via tubing **20**.

FIG. 8 is a top view of a deployment layout of a residence pod system in a body of water W that may represent a pond or lake. In the specific inventive embodiment shown, five individual mechanical basins **90** are positioned along the perimeter of the body of water W and supplies both air and biosource via flexible air tubes **34** and tubing **20**, respectively to the five residence pods **10**. The five individual mechanical basins **90** also supply air via flexible air tubes **34** to the residence air pods **50** associated with each mechanical basin **90**. Groups of tethered residence nano pods **60** are formed with connector cables **72**, and an individual group of residence nano pods **60** are tethered to each of the five mechanical basins **90** with connector cable **72**.

FIG. 9 illustrates an inventive embodiment of a residence pod **100** with one or more time release containers **102**

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holding beneficial bacteria that are distributed via the water flow produced by the bubbles generated by the air diffusers **38**. The use of time release containers eliminates the need for a peristaltic metering pump to supply biosource to the pod.

FIG. 10 illustrates another inventive embodiment of a residence pod **10'** with a main housing **12'**, a base **40'**, and an open water intake **30'** in which primed numbers have the meaning ascribed to the corresponding base number. A deflector lid **103** in a circular form directs flow all around and comes off to provide easy access to remove and replace a biosource cartridge. In some inventive embodiments, the deflector lid **103** includes an eyelet **104** to facilitate usage for mooring of other components or the lift removal of the unit. In still other embodiments, a weighted bottom mount base **106** is provided that is coupled to base **40'** or used in place thereof. The weighted base being formed of cement, metals such as iron, or composites. In some inventive embodiments, the base **106** includes at least one eyelet **108**.

FIG. 11 is a perspective view a cartridge **110** adapted to contain bacteria grow media **64** and in some inventive embodiments has a grip cutout **112** and apertures **114** for water flow therethrough. The cartridge **110** is adapted to fit within complementary openings in the residence pod **10** or **10'**. Biosource cartridges **110** are used in place of, or in combination with a reservoir **92** and are amenable to submerged placement and subject to a replacement schedule (annually, bi-annually, etc. . . .). In instances when the biosource is slow release bacteria, the need for liquid bacteria dosing is reduced as well as nutrition therefore and allows for the end users to install and buy recharge kits.

FIG. 12A is a side view of a circular array **120** of media to support bacteria grow media **64**. The constituent cartridges **122** are bundled with an open mesh wire **124** amenable to submersion. FIG. 12B is a top view of the array **120**. A central hole **126** is provide in some inventive embodiments to provide for seeding biosource material. The array **120** is modular and can be stacked with other such arrays **120'** to adjust the volume of bacteria grow media present, as shown in FIG. 12A. While only a single additional array **120'** is shown, it is appreciated that any such number can be stacked. While depicted with three annular rings of cartridges, it is appreciated that from 1 to 10 such rings of cartridges are readily provided.

FIGS. 13A and B illustrates other inventive embodiment of a residence stringer pod, shown generally at **80'** and **80''** that have weighted bases **82'** and **82''**, respectively. Weighted base **82'** is formed as a bag with a synched rope **128** extending from the weighted base **82'** in which primed numbers have the meaning ascribed to the corresponding base number. Weighted base **82''** is cast metal, cement, or aggregate. A variable length extension line **84'** has a proximal end secured to the weighted base and a distal end connected to a float **88'**. One or more media support cartridges **122** or short versions thereof **122'** are concentrically attached along the length of the extension line **84**.

FIGS. 14A and 14B illustrates a partial cutaway view of a residence pod **10'** with a main housing **12'**, a base **40'**, and an open water intake **30'**. Three of the arrays **120** are stacked therein. A cartridge **110** containing a biosource is adapted to seat within the aperture **126**. The deflector lid **103** in a circular form directs flow all around and comes off to provide easy access to remove and replace a biosource cartridge. A mounting plate **131** is provided to facilitate securement. Other reference numerals have the meaning ascribed thereto with respect to previously detailed drawings.

While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the described embodiments in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient roadmap for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes may be made in the function and arrangement of elements without departing from the scope as set forth in the appended claims and the legal equivalents thereof.

The invention claimed is:

1. A modular system for aeration and biosource augmentation of a body of water, said system comprising:

one or more air compressors located outside the body of water;

one or more residence air pods comprising:

a housing with an open top and a flow chamber;

a base connected to a lower portion of said housing and an open water intake between said base and said housing;

a series of bacterial growth media disks held in said flow chamber by a set of hanger rods in said flow chamber, said series of bacterial growth media disks spaced along an axis of each hanger rod of the set of hanger rods; and

an air distribution tube having an inlet connected to the air compressor via a set of tubes and a set of outlets connected to a set of air diffusers positioned below said flow chamber;

said one or more residence air pods configured to be submergible in the body of water, each of the one or more residence air pods in fluid communication with one air compressor of said one or more air compressors via said set of tubes, and where the one or more residence air pods are further configured to perform one or more of: providing aeration to the body of water, hosting beneficial bacteria, providing beneficial bacteria to the body of water, and delivering oxygen; and one or more residence nano pods configured to be submergible in the body of water and configured for hosting beneficial biosource; and

wherein the beneficial biosource is selected from the group consisting of microorganisms, insect larvae, enzymes, nutrients not including phosphates to promote bacterial growth, or a combination thereof.

2. The system of claim 1 wherein said one or more residence nano pods further comprise:

a base tray that supports an array of bacterial growth media;

a netting attached to the base tray that encapsulates the array of bacterial growth media; and

a mechanical fastener secured to said base tray to provide an attachment point for a connector cable.

3. The system of claim 2 wherein said array of bacterial growth media are stacked in columns at least two high and are separated by a set of spacers.

4. The system of claim 1 further comprising a residence stringer pod, said residence stringer pod further comprising: a weighted base with a top eye bolt extending upward from the center of the weighted base;

a variable length extension line with a proximal end secured to the top eye bolt and a distal end connected to a float that pulls up on the extension line toward a surface waterline when the float is submerged under the waterline of a body of water in which the residence stringer pod is placed; and

one or more media support plates each holding a bacterial growth media that are concentrically attached along a length of the extension line.

5. The system of claim 4 further comprising a second mechanical fastener on said weighted base that provides a connection point for a connector cable; and

wherein a plurality of residence stringer pods may be joined as a string of residence stringer pods.

6. The system of claim 1 wherein said biosource is bacteria provided in a cartridge in at least one of said residence air pods.

7. The system of claim 6 wherein said cartridge contains bacteria growth media and has a grip cutout and apertures for water flow therethrough.

8. The system of claim 7 wherein said cartridge is replaceable by end users.

9. The system of claim 1 further comprising a modular array of bacterial growth media in a set of cartridges bundled together with an open mesh wire that form a bundle amenable to submersion in the body of water.

10. The system of claim 9 wherein the modular array is stackable.

11. The system of claim 9 wherein the array of cartridges is arranged in annular rings with from 1 to 10 rings present in the array.

12. The system of claim 1 further comprising:

one or more basins comprising an air compressor, a liquid biosource reservoir, and a peristaltic metering pump to supply doses of the beneficial biosource from the liquid biosource reservoir, the one or more basins located outside the body of water; and

one or more residence pods comprising a housing with an open top and a partition within said housing that separates a grow chamber from a flow chamber, the one or more residence pods configured to be submerged in the body of water, to aerate the body of water, to be in fluid communication with the basin air compressor via a second set of tubes, to receive into the grow chamber beneficial biosource from the liquid reservoir via the second set of tubes, and perform at least one of: dose beneficial biosource, host beneficial biosource, or provide beneficial biosource to the body of water.

13. The system of claim 12 wherein the peristaltic metering pump is adapted to supply doses of the beneficial biosource at set intervals.

14. The system of claim 12 wherein said one or more residence pods further comprise:

a base connected to a lower portion of said housing and an open water intake between said base and said housing;

a series of bacterial growth media held by a set of hanger rods in said grow chamber;

an air distribution tube having an inlet connected to the air compressor in said basin via a first tube and a set of outlets connected to a set of air diffusers positioned below said grow chamber and said flow chamber; and

a dose receptacle that is connected to the peristaltic metering pump in said basin via a second tube that supplies a liquid of beneficial biosource to a bacterial cloth growth media supported by a screen positioned at said water intake below said grow chamber.

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15. The system of claim **14** wherein said series of bacterial growth media are a set of disks spaced along an axis of each of the hanger rods.

16. The system of claim **12** further comprising one or more replaceable biosource cartridges, where said housing of said residence pod has a circular form with a removable circular deflector lid to facilitate replacement of spent biosource cartridges; and

wherein said biosource cartridges are used in place of, or in combination with said liquid biosource supplied by the biosource reservoir by said peristaltic metering pump.

17. A method of using the system of claim **1** to prevent or eliminate an algal bloom in the body of water comprising: positioning said one or more air compressors along the outside of the water body;

connecting said one or more residence air pods to said one or more air compressors with the set of tubes for supplying air; and

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submerging said one or more residence air pods and said one or more residence nano pods in the body of water; and

wherein when there are more than one of said residence nano pods, joining the two or more of said residence nano pods with a set of connector cables to form a string of said residence nano pods prior to submerging said residence nano pods.

18. The method of claim **17** further comprising deploying one or more of a residence stringer pods and adjusting a variable length extension line extending from each of a weighted base of said one or more residence stringer pods to extend one or more media support plates each holding bacterial growth media that are concentrically attached along the length of the extension line above muck or sediment in the water body.

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